NON-MEASUREMENT SENSITIVE

MIL-T-29187B(YD)
27 June 1988
SUPERSEDING
MIL-T-29187A(YD)
20 May 1983

MILITARY SPECIFICATION

TRUCK, FIREFIGHTING: 100 FOOT REAR MOUNT AERIAL LADDER 6 BY 4, DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Naval Facilities Engineering Command (YD), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers commercial, aerial ladder fire truck for use essentially in rescue operations and combating structural fires. The truck shall be of the 6 by 4 configuration having a minimum 46,000 lb gvw rating, diesel-engine-driven, having a five person fixed canopy type cab (NON-TILT), and be fitted with a rear mounted aerial ladder not less than 100 feet in length.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4210

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SPECIFICATIONS

FEDERAL

TT-E-529 - Enamel, Alkyd, Semigloss

MILITARY

MIL-T-704 - Treatment and Painting of Materiel.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595 - Colors.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, and Trailer Dollies; Preservation and Packaging of.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.

(Copies of specifications, standards, handbooks, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other Government documents and publications. The following other Government documents, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise

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specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36/A36M - Steel Structural.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

1901 - Automotive Fire Apparatus.
1904 - Fire Department Aerial Ladder and Elevating Platforms.
1931 - Fire Department Ground Ladders.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

* 3. REQUIREMENTS

3.1 Description. Truck shall conform to the provisions of NFPA 1901, applicable to apparatus equipped with an aerial ladder, with modifications and additional requirements specified herein. The truck shall be diesel-engine-driven, front wheel steerable and rear wheel drive, equipped with a five person fixed canopy cab. A rear mounted 100-foot aerial ladder shall be furnished.

3.2 First article. When specified (see 6.1), the contractor shall furnish a fire truck for first article inspection and approval (see 4.2.1 and 6.3).

3.2.1 Inspection system requirements. Quality assurance inspection and tests necessary to substantiate product conformance to contract specifications shall be in accordance with MIL-I-45208.

3.3 Qualification. Manufacturers of fire trucks, who do not fabricate or assemble the truck chassis, shall certify to the contracting officer that the truck chassis, on which the fire apparatus is to be mounted, will be a current standard chassis model produced by a recognized motor truck manufacturer having service and parts distribution facilities in all states and territories of the United States. Certification, if required, will be made available during first article inspection.

3.4 Standard truck and accessories. Except as otherwise specified herein, truck, components, and accessories shall be the manufacturer's standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented in the manufacturer's commercially published data. Data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components.

3.5 Environmental conditions. The truck's engine shall be capable of starting and satisfactorily performing in any ambient temperature from +125 degrees (°) Fahrenheit (F) to -25°F. The truck's engine shall meet the performance requirements at any elevation from sea level to 2,000 feet at the ambient temperatures specified.

* 3.6 Performance. The truck, loaded with all specified equipment including a full crew, shall be capable of the following:

a. Ascend, both in forward and reverse gears, a grade of not less than

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30 percent. Truck to be loaded to rated gvw.

b. Accelerate on a level road from a standing start to a speed of 35 miles per hour (mph) in not more than 30 seconds, with truck loaded to rated gvw.
c. From a steady speed of 15 mph, in direct drive, on a level road, the truck shall accelerate to a speed of 35 mph in not more than 30 seconds, with the truck loaded to rated gvw.
d. The truck, loaded to rated gvw, shall attain a top speed of not less than 55 mph.
e. Service brakes shall hold the truck in position on a 30 percent grade with the truck headed in either direction. Also the service brakes shall stop the truck within 35 feet from a speed of 20 mph on a dry, smooth, level surface free from loose material.
f. Parking brake shall be capable of holding the truck in position on a 30 percent grade with the truck headed in either direction.
g. The aerial ladder shall be capable of being elevated out of bed, rotated 90°, and extended to the full height smoothly and without undue vibration in not more than 60 seconds.
h. The outriggers shall be capable of stabilizing the truck for safe aerial operation.

3.7 General design.

3.7.1 DoT Federal motor vehicle safety standards. The truck shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.7.2 Sound level. The exterior sound level produced by the truck shall not exceed the noise limits established by the EPA. The interior sound level shall be in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.94.

3.7.3 Air pollution control. Trucks shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicles Engines in effect on the date of manufacture. In addition, trucks destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

3.7.4 Curb weight. Curb weight shall include weight of chassis and cab, with all attachments, accessories and equipment, body with all specified fire-fighting equipment, full complement of fuel, lubricants, and coolant.

* 3.7.5 Gross vehicle weight. The gvw shall consist of curb weight, operator (weight computed at 175 lb), four-person crew of 700 lb, and a payload to provide not less than the specified gvw.

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* 3.7.6 Weight distribution. Suspension, axle, wheel and tire capacities shall be determined by the gvw. The payload shall be uniformly distributed over the load area. Load bearing capacities shall not be exceeded during any phase of performance required by paragraph 3.6.

3.7.7 Ratings. Vehicle rating shall be the manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. Minimum gvw shall be not less than 46,000 pounds.

3.7.8 Dimensions. The overall width shall be not more than 96 inches and the overall height not more than 130 inches. The overall length shall be not more than 36 feet.

3.7.9 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.7.9.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in an intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.8 Truck chassis.

3.8.1 Frame. The chassis frame shall provide a rigid platform and shall be of bolted construction. The chassis frame material shall be either ASTM A36/A36M steel, or high strength heat-treated alloy steel of sufficient width, depth, and thickness that will provide the required resistance bending moment to support the intended load. When operating the fully equipped and loaded ladder truck under conditions specified herein, the frame shall show no evidence of permanent deformation or distortion.

* 3.8.2 Front bumper and towing devices. A heavy duty, full width, front bumper and two tow hooks shall be provided at the front of the truck. The front bumper shall be polished, stainless steel. The front bumper shall project out at least 14 inches in front of the cab to provide a step at front of truck. A grab rail, not less than 12 inches long shall be provided on the cab front, below the windshield to facilitate personnel stepping up and onto the extended bumper. The two front tow hooks shall be mounted to the frame or

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rigid members which are attached to the frame. When access to the two front tow hooks are furnished through the front cab panel, access holes with hinged cover plates shall be provided. Two recessed, 3-inch inside diameter, towing rings, or comparable U-bolts, shall be provided at the rear of the truck and mounted directly to the chassis side rails.

3.8.3 Engine. The truck engine shall be a diesel engine having horsepower (hp) torque, and speed characteristics to meet satisfactorily all mobile performance requirements specified herein. The diesel engine shall be a liquid cooled, compression ignition, two-stroke or four-stroke cycle engine with not less than six cylinders. The truck shall meet the mobile performance requirements specified herein utilizing diesel fuel. The engine shall be fitted with the engine manufacturer's standard or recommended fuel filter system. The engine manufacturer's standard "engine start-up" procedure (engine tune-up) shall be applied before delivery of the completed fire truck.

* 3.8.3.1 Engine hour meter. An engine hour meter, having a totalizing mechanism of not less than 9,999 hours, shall be furnished for the truck chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous, trouble-free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel. The hour meter shall be the oil pressure type or a combined hour meter with tachometer.

3.8.4 Fuel system. One fuel tank having not less than 50 gallon capacity shall be furnished. The tank shall be of welded construction, shall be baffled, and shall be equipped with a readily removable strainer. The opening for filling the fuel tank shall be so located that when filling the tank, fuel will not spill on the equipment or exhaust pipe. Filler opening must be of sufficient size to permit filling of tank while engine is in operation. The tank and lines shall be amply protected against damage and excessive heat, and shall not extend any lower than chassis axle components. The fuel tank filler cap shall be the captive chained type. The fuel tank shall have a drainage outlet at the lowest part of the bottom of the tank. The fuel tank shall be provided with a fuel filter installed either at the fuel tank or in the fuel line.

3.8.5 Exhaust system The exhaust system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.83. Where necessary, exhaust pipe shall be insulated and shielded. Muffler and tailpipe shall be aluminized steel.

3.8.6 Heavy-duty cooling system. A heavy-duty cooling system shall be provided. The heavy-duty cooling system shall be of the pressurized type of sufficient capacity to maintain a cooling liquid temperature of not less than 140°F, and to provide proper cooling without overheating or loss of coolant when operated in an ambient temperature range of -25°F to +125°F during operating conditions of either stationary or mobile performance as specified. The engine coolant system shall be fitted with silicone rubber hose.

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3.8.6.1 Coolant recovery system and indicators. The coolant system shall include a de-aeration system, a surge tank, or a coolant recovery reservoir of not less than 2 quart capacity. In addition, a low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the cab instrument panel.

3.8.7 Lubrication system. The engine shall be provided with full forced- feed lubrication. The system shall include a magnetic drain plug in the crankcase and conveniently located oil fill pipe of sufficient size to permit easy filling when truck is at a standstill with engine running at full speed. Full flow type oil filter shall be furnished.

3.8.8 Governor. A governor shall be provided that will limit the engine to the maximum full load revolutions per minute (rpm) recommended by the engine manufacturer. The governor shall provide maximum effectiveness under all engine operating conditions and shall not restrict engine power output below governed full load rpm.

3.8.9 Electrical system. The truck shall be equipped with a complete 12-volt (V) negative ground electrical system. Components shall be located so as not to be adversely affected by engine heat. Wiring shall be waterproofed, color coded, or numbered properly secured, and protected at points of potential abrasion.

3.8.9.1 Alternator. An alternator shall be furnished complete with rectifier and regulator. The alternator shall have a minimum capacity 145 ampere direct current (dc) power.

3.8.9.2 Battery system. A 12-V battery system shall be furnished. A minimum of two batteries shall be furnished. Each battery in the system shall be 12 volt and of the sealed cell (maintenance free) type. Each battery shall be rated in terms of cold cranking capacity at 0°F for the engine furnished. Battery compartment(s) shall be provided in the canopy cab area, as close as possible to the engine in order to reduce battery cable length for connection. Access shall be provided for ease in service and removal. The battery cables shall be protected by use of grommets where the cables pass through metal bulkheads and shall be covered by a protection boot where the cables are attached under vehicle frame to the electrical switches or connectors. The battery compartment(s) shall be ventilated and treated against acid fumes and spillage. The batteries shall be provided with a polarized receptacle(s) for station charging. The receptacle shall be located near the batteries at the driver's entrance to the cab. The charging receptacle shall be suitably identified. The receptacle shall be of the recessed male type with a spring loaded cover and provided with a matching female plug with protected prongs. Contractors shall also provide the female plug for the receptacle. A carrying handle for the batteries shall be provided. Removable battery compartment cover(s) shall be fitted with lifting handles. Finger lift openings are not acceptable.

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3.8.9.3 Ignition. The ignition system shall be the manufacturer's standard furnished with a keyless switch and a pilot light.

* 3.8.9.4 Lighting system. The truck shall be provided with a complete 12-volt negative ground starting, lighting, and ignition system. Electrical equipment, including all truck lights and wiring shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.19, 393.20, 393.22, 393.24, 393.25, 393.27 through 393.31, and 393.33, except as specified herein. Reflectors shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.26. Directional signal lamps shall be in accordance with SAE J588 and shall be arrow shaped, amber in color front and rear. The operating unit shall be in accordance with SAE J589, class A. Hazard warning signal switch, in accordance with SAE J910, which flashes all turn signal lamps simultaneously, shall be furnished. Wiring shall be weatherproof, resistant to moisture and fungus, and color coded or numbered for ready identification. Wiring identification shall be consistent throughout the truck. The truck chassis electrical circuits and the truck body electrical circuits shall be separate circuits. The truck body electrical circuits, excluding truck chassis electrical circuits, shall be provided with overload protective devices of the automatic or manual reset, overload-circuit-breaker type. Two spare single pole, 15 ampere breakers shall be furnished for future radio use. All lamps installed in rear of body shall be flush mounted and so located to not interfere with firefighters climbing up and down rear steps to ladder platform. All lights specified herein shall have chromium plated metal housings.

3.8.9.4.1 Rear marker lamps. The rear marker lamps in accordance with DoT Federal Motor Carrier Safety Regulations 393.12b shall be located as follows:

a. The three identification lamps located on centerline of body shall be mounted at the highest location on rear of body as practicable.
b. Side rear marker lights shall be located on rear edge of body on horizontal centerline of the three identification lamps. Additional side rear marker lamps shall be mounted in recessed areas, one in each rear corner of rear bumper/step.

* 3.8.9.4.2 Service lighting system. In addition to the standard running lights specified in 3.8.9.4, the following lights shall be furnished.

a. Cab dome lights shall be furnished in the driver's compartment and in the rear canopy area.
b. Two automotive type, 6-inch sealed beam, roof type spotlights shall be mounted to the roof of the cab. One spotlight shall be located near the left windshield post and one shall be located near the right windshield post. Controls shall be extended inside the cab for ON-OFF control and movement of lamp housing through a horizontal angle of at least 270°, and a vertical angle of at least 90° above and 35° below horizontal.

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c. Two backup lights, flush mounted at rear of body. The lights shall be automatically activated when vehicle is in reverse gear operation. The lights shall be not less than 4-inch diameter.
d. Suitable lighting at the base of the aerial ladder to illuminate approaches to the ladder control station and also two spotlights to illuminate the ladder in any position of operation.
e. Suitable lighting shall be provided for all instruments, gages, ladder panel controls, and outrigger controls.
f. Compartments specified in 3.11 shall each have one or more lights for illumination. Each light shall be similar or equal to 2-1/2 inch model manufactured by the Griffin Lamp Company. A master control switch shall be mounted in cab instrument panel with flashing pilot light to indicate "ON position" and identified as to function.
g. Four Collins FX-12, or equal, 12 V Spotlights shall be mounted, two on the base section, and two on the top of the ladder section. Each light shall have an ON/OFF switch on the light. Lights shall be mounted below handrail height so as not to increase the overall height of the unit/truck. A master switch for these spotlights shall be provided on the turntable console.

* 3.8.9.5 Emergency lighting system. The truck shall be equipped with an emergency lighting system as follows:

a. Two revolving 12-volt warning lights shall be provided and mounted, with reinforcement at point of attachment. Each warning light shall have a clear dome, shall have four sealed beam lamps each having a minimum of 35,000 candlepower and a minimum flash rate exceeding 80 flashes per minute. Two of the lamps shall be clear and two shall be red. The lamps shall be installed in alternating colors. The minimum warning light dimensions shall be: 9 inches high, 9-inch diameter 5-inch skirt (base). The warning lights shall be mounted on the cab roof, one on each side, located approximately 6 inches inboard from the edge of the cab roof and approximately midway between the windshield and the forward edge of the canopy roof.
b. Two revolving warning beacon lights similar and equal to KD brand lamps style KD886, having a rotating flash rate exceeding 80 flashes per minute. The beacons shall be mounted on top of rear body cabinets, one on each side, in a location which does not interfere with firefighters climbing to and from ladder base.
c. Two red flashing warning lights shall be mounted on the front of the truck, installed on a horizontal plane through the centerline of the turn signals. The warning lights shall be approximately
7 inches in diameter similar and equal to Weldon model 1020 having a minimum 200 cp. A heavy-duty flasher unit shall be furnished which provides for a combined alternating flash rate of not less than 80 flashes per minute.

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d. Side marker lamp. A series of red side marker lamps, mounted to bottom edge of compartments (left and right), shall be furnished. One or more shall be furnished behind the rear wheel housing. Not less than five shall be furnished forward of rear wheel housings spaced or not greater than 30-inch centers. The lamps shall be similar or equal to Grote model 46352 with red lens.

e. A single control switch to operate the front and rear revolving warning lights, front flashing lights and red side marker lamps, shall be mounted in the instrument panel, accessible to the driver. A red pilot light, indicating the "ON" position, located in the instrument panel, near the control, shall be provided. The switch shall be identified as to function.

* 3.8.9.6 Siren. An electric, motor driven siren equal to Federal Q2b type, pedestal mounted on the driver's side of the bumper extension shall be furnished. The siren shall have floor mounted activator devices (driver and passenger positions). A siren brake control shall be mounted at the driver's side of the instrument/dash panel, identified, and in a location readily visible to the driver.

* 3.8.9.6.1 Radio interference suppression. The fire truck suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the truck chassis.

* 3.8.9.7 Horn. An electric horn of manufacturer's standard type shall be furnished. Also, two chrome plated air horns of the studder tone type shall be provided on the cab roof, one horn on each side of the roof. The air horns shall be controlled from one floor mounted, foot operated control located on the fire officer's side of the cab and a lanyard (pull type) control from the overhead center of the cab.

* 3.8.10 Transmission. An automatic or semiautomatic transmission shall be provided. The transmission shall include a hydraulic torque converter and not less than five forward gear ratios. The gear ratios shall be of the wide ratio type which provides a first gear ratio of not less than 7.5. The shift control shall provide for driver selection of individual gears or ranges of gears. The transmission shall be provided with SAE J704 power takeoff (pto) opening(s). Net torque capacity of the transmission shall be not less than 480 pound-feet and a net input power rating of not less than 250 hp. The transmission shall have a torque converter lockup in the two highest forward gears. Torque converter multiplication factor shall not be used to establish first gear ratio of not less than 7.5.

3.8.10.1 Transmission capacity. The torque input capacity of the converter transmission assembly shall be in excess of the net torque output of the engine, and the maximum speed output of the engine shall not exceed the rated input speed of the transmission. The converter with transmission design shall acknowledge net flywheel hp.

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3.8.10.2 Torque converter. The torque converter shall be heavy duty having a multiplication factor of not less than 2 to 1 at stall. The torque converter shall be equipped with a heat exchanger. The torque converter transmission combination shall be provided with an oil filter on the low pressure side of the lubricant fluid level.

3.8.10.3 Converter gage. A torque converter temperature gage, suitably mounted on the cab instrument panel, complete with flashing warning light that will be clearly visible to the operator when the temperature of the converter fluid exceeds the unsafe temperature, shall be provided. The signal light shall be suitably identified with a permanent metal plate. An additional audible device shall be provided which will actuate the horn when the converter fluid temperature exceeds 265°F.

3.8.11 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

* 3.8.12 Suspension system. Truck shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with truck loaded to rated gvw.

3.8.13 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with truck loaded to rated gvw.

* 3.8.14 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.42(b), 393.43, 393.45 through 393.52, and as specified in the current FMVSS-121 standards.

3.8.14.1 Service brakes. The service brakes shall be of the air application type. Air filters shall prevent brake system contamination. Brake operation shall not be subject to excessive fading.

* 3.8.14.1.1 Brake system compressor. The capacity of the air compressor shall be sufficient for the intended service but shall be not less than 15.5 cubic feet per minute. The air compressor shall be equipped with a governor and the compressor intake shall be properly connected to an air cleaner. An air dryer, Bendix Model AD-4 or equal, shall be furnished and installed to remove moisture from the air. The moisture removed from the air by the air drying system shall be disposed of automatically and the moisture collector shall be winterized to prevent freezing.

* 3.8.14.1.2 Brake system air storage reservoirs. The air storage reservoir shall have two air gauges, with indicator lights and buzzer alarm located inside the operator's cab. The system shall be of the quick-release buildup valve with the air compressor connected to a small reservoir which is pressure relieved into the larger reservoirs when the minimum pressure is reached for optimum operation of the brake system. The time required to

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reach this pressure from zero pressure at 50 percent of engine governed speed shall not exceed 30 seconds. If other than brake air supply requirements are to be supplied from the reservoirs, a priority system shall be installed to assure that an adequate supply of air is always available to operate the brake system. All reservoirs shall provide a minimum of 4500 cubic inches and have manually operated, conveniently located, water drain valves.

* 3.8.14.2 Parking brakes. Parking brakes, with independent operation and control from the air service brake system, shall be provided. The parking brakes shall be spring type on all rear wheels and have an indicator light and control valve located within reach of the driver. The parking brakes shall be capable of meeting the performance requirements specified in 3.6.

3.8.14.3 Air brake emergency system. An emergency brake system shall be furnished. The system shall be capable of being manually applied by the operator seated in the driver's seat. A manual or mechanical means to release the brakes shall be provided when compressed air is not available to release the brakes. The emergency system shall also serve as a parking brake and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.41 and 393.52.

* 3.8.14.4 Backup warning device. The truck shall be provided with an automatically activated, audible, pulsating signaling warning device, mechanical or electrical, to caution personnel when the truck is in reverse gear operation. The backup alarm shall have a noise level of at least 100 decibles at a distance of 10 feet.

* 3.8.15 Wheels, rims, tires, and tubes. The truck shall be equipped with single front and dual rear wheels. Tire and rim ratings shall conform to TRA recommendations for the type and size of tires furnished. Tires shall be tube or tubeless type with highway tread. Tires shall be steel belted radials. The rated individual tire capacity at 50 mph highway conditions shall be not less than the maximum wheel loading imposed with truck loaded to rated gvw. Wheel and tire assemblies shall be dynamically balanced at simulated speeds up to 55 mph. Disc type wheels shall be provided. Each wheel assembly shall be balanced individually.

3.8.15.1 Inner tubes. For tube type tires, inner tubes shall be of heavy-duty type and shall be of the proper size for the tires furinished. Tire flaps shall be provided for tube-type tires in accordance with TRA recommendations.

3.8.15.2 Spare wheel. A spare front wheel with tire shall be furnished and stowed on the truck for shipment.

3.8.16 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed

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and located to permit easy entrance and exit of operator, to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instrument controls and wiring thereto shall be incorporated into modular design panels and shall be removable by quick disconnects for ease of maintenance.

3.8.17 Steering mechanism. The truck shall be equipped with a power assist steering mechanism of hydraulic type. The mechanism shall permit manual steering in the event of failure of the hydraulic system, and shall be properly effective at all truck speeds. The system shall be furnished complete with all necessary piping, valves, and accessories. The steering mechanism shall provide for controlling the direction of the fully loaded truck under all operating conditions. The cramping angle shall be not less than 30° right and left.

* 3.8.18 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished and suitably identified:

a. Keyless ignition switch with pilot light.
b. Ammeter gage or charging indicator.
c. Fuel gage.
d. Oil pressure gage.
e. Engine temperature gage.
f. Speedometer with recording odometer.
g. Tachometer.
h. Dual sunvisors.
i. Hour meter.
j. Service brake air pressure gage.
k. Brake quick buildup gage.
l. Torque converter temperature gage.

* 3.8.19 Heater and defroster. Enclosed portion of the cab shall be equipped with a hot water type heater having fresh air intakes, discharge outlets to floor, and windshield defroster louvers. Heaters shall be equipped with a 12V, electric-motor-driven, two-speed blower controlled by a dash mounted, three-position switch. The location of the heater shall not restrict or interfer with personnel when entering or exiting the cab or while seated.

3.8.19.1 Defrosting, anti-icing, and ventilating. A sufficient defrosting, anti-icing, and ventilating system shall be provided in the cab. This system shall have the capacity to effectively prevent formation of frost in the interior surface of all transparent areas of the cab and to prevent formation of ice on the exterior of the vehicle windshield. The system shall also maintain clear vision over at least 75 percent of the area of all windows of the cab interior.

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3.8.19.2 Controls. Variable controls shall be provided on air blowers and at least high output setting shall be furnished on the cab heating system controls. Adequate operating instructions shall be provided at heater system control point. Cab heater and defroster system control shall be mounted on the instrument panel.

3.8.20 Rearview mirrors. Outside rearview mirrors, each with at least two supporting arms or heavy-duty spring loaded ratchet type mounting brackets, shall be mounted on each side of the cab. Combination (flat and convex) mirrors, enclosed in a common housing, shall be furnished. Each combination mirror shall have at least 50 square inches of reflective area and a convex surface having at least 20 square inches of reflective area.

* 3.8.21 Windshield wipers. Heavy-duty electric operated windshield wipers shall be mounted to facilitate clear vision. The wipers shall be of such capacity and coverage as to readily cleanse the windshield of rain, snow, and ice when used in conjunction with the defrosting system or windshield washer mechanism. The windshield wipers shall meet the requirements specified in SAE J198. Wiper assemblies shall include features to permit speed control and return of the wiper blades to a retracted position out of the line of vision when not in use.

3.8.21.1 Windshield washers. Power operated windshield washers, equipped with a minimum of 1 gallon capacity plastic container for cleaning fluid shall be furnished.

* 3.9 Cab. The cab shall be a five-person fixed CANOPY TYPE (NON-TILT). The canopy portion of the cab shall allow for readily accessible engine maintenance and removal. The engine hood (cover) in the canopy cab section shall be insulated against heat and noise, but shall allow for adequate venting away from seated personnel.

* 3.9.1 Cab construction. The cab shall be of all aluminum reinforced, welded construction. The cab interior width dimension shall be not less than 88 inches. With the driver's seat in the rear-most position, cab width shall be measured at the floor on a transverse line through a vertical plane, 6 inches forward of the driver's seat cushion. Aluminum sheet metal used in construction of the cab shall be at least 0.125-inch thick. The cab shall provide unobstructed seating and head room for a minimum of five persons fully clothed in firefighting gear. In the forward section of the cab, seats shall be provided for two passengers and a driver, and in the rear, under the canopy section, two jump seats shall be furnished. The two passenger seats in the forward cab will be jump seats designed to accommodate an air pack as described in 3.9.2. The forward enclosed section of the canopy type cab shall be provided with a door on each side of the fire truck. The cab doors shall be ample size for easy access to the forward cab and shall be equipped with roll-down windows. The doors shall be provided with locks and with at least the front curbside door equipped with an external key-operated lock. Grab handles shall be provided on each side of the fire truck at each cab

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door and inside the cab at each end of the instrument panel. Additionally, two grab handles shall be provided at each canopy entrance of each side of the truck. The outside grab handles shall be of a nontwist design, polished stainless steel, and shall be not less than 1-1/4 inches in diameter. A space shall be provided between grab handle and cab body for grasping with heavy gloved hand. A step shall be provided at each cab door and at each canopy entrance. Each step shall have an aluminum, nonslip tread surface. Each step shall support, without permanent deformation, the weight of a 250 lb firefighter in full firefighting gear. Each step shall be of maximum width and depth to permit safe entry and exit for firefighter wearing fire-fighting foot gear. Safety glass shall be provided throughout the cab. Windows between the forward cab and canopy shall be sliding glass with finger pulls and equipped with a latching device. The channel for the sliding window shall be lined to reduce rattles. A dome light shall be provided in the forward cab, two step lights shall be provided, and two dome/map lights shall be provided in the rear canopy section, the forward dome light and two step lights shall be controlled by door and manual switches. Rubber boots shall be provided for all control floor board openings. Dual sun visors shall be provided. Cab floors and walkways shall be of heavy gage aluminum diamond plate. A full circular, wheel well liner shall be furnished to protect against dirt and water entry. The forward cab shall meet the interior noise criteria of DoT Federal Motor Carrier Safety Regulations, section 393.94.

* 3.9.2 Seats. All seats shall be upholstered over deep air foam cushions with first grade black vinyl covering, suitable for fire service. All seats shall be equipped with an adjustable headrest restraint. The driver's seat shall be a heavy duty, four way adjustable air ride bucket type seat similar or equal to Anchorlok brand. The fire officer/passengers and the two rear facing under canopy seats (jump seats) shall be Ziamatic Quick-Seat TM or equal. Each jump seat shall have a padded lower cushion of suitable width and depth to provide ample seating for firefighters wearing turn out gear and back strapped air packs. The back of each jump seat shall have an air breathing apparatus walk away bracket, Ziamatic Model U45F or equal. The bracket shall provide for mounting of the air breathing apparatus in an upright position. A series of predrilled holes shall be provided in jump seats back for vertical adjustment of walk away brackets. Each jump seat back shall be provided with padded flanges to each side and forward of the breathing apparatus. Seat belts shall be provided for the driver and passengers/jump seats of sufficient length to accommodate firefighters wearing turn out gear. The rearward facing jump seats shall provide additional storage compartment(s) if available in manufacturer's basic design. The right rear facing jump seat shall have at least 4-inch spacing between seat and cab exterior wall and the left rear facing seat, (behind driver) shall have at least 4-inch spacing between seat and engine enclosure. A space of at least 18 inches between bottom of jump seat and canopy decking shall be provided.

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3.10 Rear platform and running boards. Rear platform and running boards, fabricated from heavy-gage aluminum or steel "diamondette," shall be furnished. Aluminum or steel kick plates shall be furnished at cab and canopy entrances and forward edge of rear platform. The rear platform shall have a load carrying capacity of not less than 700 lb. The rear platform shall be full width of body and not less than 15 inches from rear edge to rear of the body. The corners of the platform shall be rounded or formed at a 45° angle with recess for mounting of rear side lower marker lamps. The side running boards shall extend from rear of cab to forward end of body and shall have a load carrying capacity of not less than 350 lb. The rear platform and side running boards shall be not greater than 22 inches from ground level with truck loaded to gvw.

* 3.11 Compartments. The maximum cubic feet of compartment space shall be furnished on each side of the truck. Manufacturer's shall provide diagram depicting compartment design and layout.

* 3.11.1 Compartment construction. All compartments shall be fabricated from aluminum and be at least .09 inches thick. Compartment floors will be integral with the compartment, be constructed of aluminum and be the sweep out type. Compartment door openings are to be framed by flanging edges to form a channel. Compartment doors will be flush mounted with double panel construction of aluminum. Outer panel door edges will be framed in a channel design to contain an inner pan. The interior of double panel doors will be coated to inhibit corrosion. A heavy duty automotive type extruded rubber moulding shall be installed on interior door pan and exterior edges to ensure a weatherproof compartment. All compartment doors will have polished stainless steel continuous piano type hinges. Hinge pin diameter will be a minimum of 3/16 inches and stainless steel material. Doors will be equipped with Eberhard M206, or equal, two-point latches that close on a single striker. Door handles will be the bent "D" ring, polished stainless steel type. The free door, of double door equipped compartments will have a latch also. Cleveland type door holders are to be installed on each door. Switches for automatic compartment lights are to be installed at the hinge side of door. Drip moulding will be installed above all compartment door openings. An aluminum treadplate walkway over the fender side compartments shall have rolled over edges (FRONT-REAR-OUTSIDE EDGES) and formed out to provide drip protection, with inner body edge being rolled upward a minimum 1.5 inches. A compartment shall be provided (MID-SHIP MOUNTED) and contain a cascade system for recharging self contained breathing apparatus. The cascade system shall meet NIOSH and American National Standards Institute standards (ANSI) with the capacity to recharge a minimum of 25 thirty-minute air bottles. The unit shall have a fill station on the left side, consisting of the following: individual shut off valves for each cylinder, individual pressure gauges for each cylinder, one master shut off gauge for the fill line, and one six foot hose for filling the air bottles.

* 3.12 Aerial ladder. The aerial ladder should consist of a minimum of four sections and extend to a nominal working height of not less than 100 feet above the ground. Ladder design should incorporate a minimum safety

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factor of three to one, based on the live load. A one and one-half to one stability factor in compliance with the OSHA and the (ANSI) is required. Ladder construction should be of welded, steel tubing with a minimum of 46,000 pounds-force per square inch high-strength. Each section of the ladder shall be trussed diagonally, vertically and horizontally and should be reinforced at critical points to provide extra rigidity. All ladder rungs are to be round and welded to each section using "K" bracing. All rungs are to be covered with serrated replaceable heavy-duty rubber sheaths, which are to be glued and clamped securely to the rungs. Inside width dimensions of the four ladder sections shall permit the passing of personnel on the ladder and be no less than 36 inches wide at the base section and no less than 20 inches wide on the fly section. The height of the handrails above the rungs shall be no less than 26 inches on the base section, 23 inches on the inner section, 19 inches on the outer section and 15 inches on the fly section. Folding steps will be provided for the operator's footing while at the tip of the fly section. A ladder rest shall be provided to support the ladder while in the travel position. Mounting brackets shall be provided on the inside of the fly section for an axe and a 10-foot pike pole. A telescopic water pipe system shall be provided on the ladder and a monitor with nozzle equal to the Akron model 3478. The piping sizes shall be a minimum of 4.5 inch outside diameter in the base section, 4 inch outside diameter in the inner mid-section, 3.5 inch outside diameter in the outer mid-section and 3 inch outside diameter in the fly section. A pipe will be provided to connect the water supply on one end and to a water swivel coupling that will allow the ladder to rotate 360° continuously while pumping water. A pipe will be routed through the hydraulic and electric swivels to the heel pin swivel to allow water to flow to the ladder pipe while elevating the ladder -6° to +75°.

* 3.12.1 Aerial ladder raising device. A minimum of two double-acting lift cylinders shall provide elevation from at least -6° to +75° above horizontal. The cylinders shall be equipped with holding valves that will prevent the unit from dropping, even if a charged hydraulic line should fail anywhere in the system. A fully hydraulically powered extension and retraction system using two hydraulic cylinders, with each cylinder being able to operate the ladder should one of the cylinders fail, shall be provided. The aerial ladder capacity, in an unsupported position, based on 360° rotation, and full extension from -6° to +75° shall be a minimum of 400 lbs at the tip. There shall be two gauges at the control console that show the system hydraulic pressure and the ladder load. An alarm shall be provided that will have an audible and visual warning that will activate when the ladder load capacity is exceeded. The aerial water system shall be able to pump 1000 GPM while fully extended and 90° to each side of center with a 400 lb load at the tip. Adequate hand clearance shall be provided between the sides of the ladder and moving cylinder(s) or other moving parts, or suitable hand guards shall be provided to prevent injury. A calibrated safety valve in the return oil passage of the cylinder(s) shall automatically control the lowering speed of the ladder. The ladder control levers for all ladder operations shall be grouped in an operator's control pedestal located

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on the platform. The aerial ladder shall be capable of being elevated from the ladder rest, rotated 90°, and extended to the full height smoothly and without undue vibration in 60 seconds or less.

* 3.12.2 Aerial ladder platform and turntable. A sturdy operating platform, built of heavy-duty aluminum diamond plate, shall be provided. The platform deck shall provide an ample unobstructed area for all rescue and service operations. The turnable is to be mounted at the rear of the truck with the ladder extending forward. Steps and handholds or rails shall be provided to reach the turnable from the rear of the truck on both sides. Steps shall be illuminated. The turntable platform shall be provided with a safety railing of polished stainless steel tubing 1-1/4 inch in diameter at least 36 inches high located on each side of the turntable platform. The turntable treadplate shall have a downward facing of at least 2 inches. Anti-skid strips shall be applied at the base of the ladder on the tread-plate. The turntable shall be well lighted by lights activated from the aerial master switch. Access/ clearance, as specified in 3.17, shall be provided for the turntable mounting bolts so the correct torque can be checked in accordance with procedures specified in NFPA 1904.

3.12.2.1 Aerial ladder two-way communication system. A two-way intercom system shall be provided at the aerial ladder platform operator's position and the outer fly section to allow continuous communication between ladder operator and firefighters. The intercom unit shall be similar or equal to model IC-9 manufactured by RATELCO Inc. The system shall be powered by chassis battery system, shall have at least 18 watt output and shall be weatherproof design. The firefighter aloft at the fly section shll have no controls to operate ("Hands Off" communication).

3.12.3 Aerial ladder control, locking, and safety devices. Clearly marked suitably lighted, and foolproof deadman control, locking and safety devices shall be provided to perform the following:

a. Immobilize the truck.
b. Transfer power application to the ladder mechanism.
c. Lock in and release the ladder from its bed.
d. Stabilize the truck.
e. Control elevation, rotation, and extension of the ladder.
f. Indicate the angle of elevation, rotation, and extension of the ladder.
g. Visual indication of the safe load at any angle or extension of the ladder.
h. Start and control the engine from the operator's position.
i. Lock control for ladder rung locks or pawls.
j. Provide auxiliary power (see 3.12.3.7) operation of the ladder in the event of power failure.

3.12.3.1 Immobilization controls. Controls for immobilizing the vehicle shall be installed at the driver's position in order to engage all

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necessary parking brake systems before the driver leaves the cab. In addition, four wheel chocks are to be provided to be positioned by driver before aerial ladder operation.

* 3.12.3.2 Ladder bed lock. A lock shall be provided which will prevent action and movement of the aerial ladder in its bed when the vehicle is in motion. The lock release or control shall be located at the operator's position at the turntable or may be automatically performed when actuating the controls required. An interlocking system of limit switches shall be provided to prevent the lifting of the aerial ladder from the bed lock position until the outriggers have been deployed to the load supporting position and also to prevent any movement of the outriggers while the aerial ladder is raised from the bed lock position.

3.12.3.3 Elevation controls. A control for elevating the ladder out of its bed to the maximum angle of elevation and returning it shall be provided at the operator's position. This control shall be arranged to permit the operator to regulate the speed of elevation within safe limits determined by the manufacturer. A manual lock valve or other positive locking device shall be provided so that the desired angle of elevation can be maintained indefinitely without dependence upon engine power.

* 3.12.3.4 Turntable controls. Controls for rotating the turntable 360° in either direction shall be arranged to permit the operator to regulate the speed of rotation within limits determined by the manufacturer. The turntable shall be fitted with strong friction or other positive locking device so as to retain it in any desired position. Suitable indicators, clearly visible at night, shall be provided to facilitate correct alignment of the turntable with the ladder bed. A back-up manual system shall be provided to operate the rotation of the turntable in the event of a loss of power. The control console shall be well lighted with all control switches clearly identified and easily accessible. The control switches for each ladder section shall allow individual and/or simultaneous operation for extending and retracting the ladder sections. An automatic throttle control shall be provided to increase the engine idle speed to a pre-set rpm when the ladder sections are being extended, retracted, raised or lowered, and return the engine to idle speed with the ladder activating system is in the off position.

3.12.3.5 Ladder extension controls. Controls shall be installed at the operator's position for extending the fly sections of the aerial ladder to any desired height between the fully retracted and fully extended positions, the same control may also be used to retract the sections. Safety provisions shall prevent damage at full retraction or extension. Provisions shall be made so that in the event of failure of extension mechanism during operation, the descent of the ladder shall be kept at a speed which will prevent damage to the ladder structure or danger to personnel. Stops shall be provided to properly align the sections without damage to the ladder or the cable mechanism when the ladder is retracted.

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* 3.12.3.6 Ladder lock controls. A lock that controls ladder rung locks or pawls shall be provided at the base of the ladder adjacent to the operator's position to prevent movement of the sections when the power is shut off or in the event of loss of pressure in the hydraulic circuits. This lock shall be arranged to use when alining rungs. The control for the pawls shall be such that the on and off position may be clearly determined by the operator without application of power when the pawls are locked, it is desirable to have an automatic device to prevent retraction of the ladder sections until pawls have been released. The above locking can be accomplished manually and/or hydraulically.

3.12.3.7 Auxiliary power system. Provisions shall be made so that an auxiliary electric motor shall power the hydraulic system for all ladder function when normal operating power source fails. If required, manual controls shall be provided which will permit releasing of the pawls for retraction of the ladder in the event the auxiliary power system is used.

3.12.3.8 Engine controls. A supplementary engine starting control shall be provided on or adjacent to the operator's control pedestal. If the engine speed is not automatically regulated or predetermined, a suitable engine throttle shall be provided but shall be governed to prevent operation of the ladder at excessive speeds and a tachometer shall be installed. The truck engine shall be provided with a fast idle device set for 1000 rpm or for the design rpm to operate the hydraulic system and shall be manually activated from the driver's position in the cab.

3.12.3.9 Angle of elevation indicator and load limit. A device, clearly visible day or night from the operator's position, shall be provided to indicate the angle of elevation of the ladder at all times. Either a load indicator or an instruction plate visible from the operator's position shall show the recommended safe load under existing conditions of ladder elevation.

3.12.3.10 Power transfer control. Controls shall be provided at the driver's position to transfer the power from the road drive to the aerial ladder mechanism.

* 3.12.3.11 Training. The manufacturer of the aerial ladder and related control systems, shall provide a minimum of 24 hours training to final destination fire department personnel in the operation and maintenance of the aerial ladder. Training to take place at site of receiving customer destination.

* 3.13 Outriggers. Hydraulic outriggers shall be provided and positioned to fully support the frame and turntable structure. The hydraulic outriggers shall be controlled at the rear of the apparatus with individual controls for each outrigger. Automatic pilot check valve shall be provided for each of the outriggers in the event of a damaged hydraulic line. An indicator light shall be furnished at the operator's control station on the turntable to signify that the outriggers have been extended. A self-actuating locking device shall

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prevent accidental raising of the outriggers while the serial ladder is elevated from stowage position. All outrigger controls shall be installed in a readily accessible, well protected location which permits operation from ground level. Each outrigger shall be provided with an automatic, positive, spring actuated lock, or an automatic pilot check valve which holds the outrigger in the retracted position without settling. The outrigger shall withdraw within the overall dimensions of the truck. Floodlights shall be provided at each outrigger station to illuminate outrigger positioning area. A readily visible leveling indicator shall be provided and properly illuminated to indicate proper stabilization of the truck to the operator when setting the outriggers. This leveling indicator shall be positioned at the rear center line of the truck. An audible alarm shall be provided on each side of the truck and will sound a warning when the outriggers are being deployed or retracted.

3.13.1 Outrigger movement. The outrigger movement span shall be not less than 145 inches.

* 3.13.2 Outrigger ground pads. Each outrigger ground pad shall have a ground contact surface area of not less than 140 square inches. Ground pads shall be of the hinge or swivel type. If the ground pad swivels in one direction only, it shall be in a plane normal to the axis of the apparatus. Additional foot plates shall be furnished to increase the ground contact area of each outrigger to not less than 320 square inches. Storage space for the foot plates shall be provided on the vehicle.

3.13.3 Outrigger warning lights. Red indicator/warning lights, suitably identified shall be furnished at the ladder operator's control pedestal to indicate that the outriggers have been extended.

3.13.4 Outrigger warning plate. A warning plate shall be provided, installed at ladder operator's position on turntable, that reads as follows: "DO NOT OPERATE AERIAL LADDER UNLESS THE OUTRIGGERS ARE DOWN AND IN THEIR LOCKED POSITION."

3.14 Hydraulic system. The hydraulic system shall consist of all components essential to the efficient operation of the aerial ladder and outriggers.

3.14.1 Hydraulic system pump. The pump which supplies the liquid for all hydraulic operations shall be the manufacturer's standard type driven by pto from the truck engine. The pto engagement lever shall be located in the driver's compartment. It shall be provided with a lock mechanism to prevent accidental engagement or disengagement, and shall be suitably marked for identification and operating procedures and, in addition, a red warning light shall be provided on the instrument panel in the cab to warn the operator the pto is engaged. The pumping system shall include an adjustable pressure relief valve or unloading valve with the range of adjustment which allows raising or lowering of the aerial ladder with the working or test loads specified herein.

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3.14.2 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid with not less than 10 percent for expansion space. The reservoir shall be provided with a filler opening and removable oil strainer and a vented filler cap, accessibly located to facilitate filling the tank from a 1 gallon container using a funnel. The vented filler cap shall have means to preclude fluid from splashing out of tank during movement of vehicle. The filler cap shall be attached to the tank by a corrosion-resistant chain to prevent loss. Provision shall be made for inspection and cleaning of interior and draining of the reservoir. The reservoir shall have a photo-sensitive, corrosion-resistant plate indicating the type of hydraulic fluid required and the maximum level in the reservoir with the aerial ladder in travel position and with the hydraulic lines and components full of fluid. A sight level or dipstick gage shall be furnished to indicate level of fluid.

3.14.3 Oil filter and strainers. A full-flow type 10-micron discharge oil filter in the return line to the reservoir shall be provided. A 100 mesh suction line strainer in the hydraulic reservoir shall be provided. Filter and suction strainer shall be easily accessible for periodic cleaning, or for replacement of the filter cartridge. Filter and strainer shall have sufficient capacity to allow the maximum rated pump discharge.

3.14.4 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, visual inspection for failures and maximum protection from damage during operation of the aerial ladder. Hydraulic lines and fittings shall be air and liquid tight and shall be installed to preclude vibration and chafing. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic fluid lines. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature, and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be furnished.

3.14.5 Hydraulic gage. An oil pressure gage shall be provided at the ladder operator's position indicating the hydraulic pressure.

3.15 Ground ladders. Aluminum ground ladders shall conform to provisions of NFPA 1931, Article 100. Extension ladders shall be equipped with the manufacturer's latest design safety locks and raising mechanisms. The length of the ladder shall be the measured length when extended and not the sums of the lengths of sections. The following ladders, in the quantity shown, shall be furnished and installed on the truck:

a. Two 35 foot, two-piece or three-piece extension ladder.
b. Two 28 foot, two-piece extension ladder.
c. One 30 foot, two-piece extension ladder.
d. One 18 foot, single ladder with folding roof hooks.
e. One 16 foot, single ladder with folding roof hooks.
f. One 10 foot, collapsible ladder.
g. One 14 foot, single ladder with folding roof hooks.

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The ladders shall be carried in a double bank arrangement and locked down by quick-acting locking devices.

3.15.1 Ground ladder storage. Ground ladders shall be stored in the truck bed under the aerial ladder. Access to the ground ladders shall be from the rear of the truck. The ladder storage space shall be boxed in to prevent entry of ground water, and road grit from the underside of the truck. Teflon or oak stripping shall be used to protect ladder bed runners against continuous rubbing and chafing during removal and installation of ladders.

* 3.16 Pike poles. The following fiberglass type pike poles shall be furnished and stowed in the truck adjacent to or along side of the ground ladder(s). If tubing is used for storage, the material shall be aluminum. In addition to the pike poles specified in items "A" through "D", the basic accessory equipment as listed in paragraph 10-4.1 of NFPA 1901 shall be provided.

a. Two - 6 foot.
b. Two - 8 foot.
c. Two - 12 foot.
d. Two - 16 foot.

3.17 Ease of maintenance. The truck shall be so designed and constructed that normal adjustments, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or disturbance of other elements of the unit. Ease of maintenance provisions shall insure operating clearances for facilitating maintenance and servicing. Where possible, intricate locking devices, controls and threaded fastenings that can be easily overtorqued by operators lacking feel through thick gloves or numbness shall be avoided. Covers or plates that must be removed for component adjustment or for parts removal shall be equipped with quick-disconnect fastenings.

3.18 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Where the use of high lubricating pressure will damage grease seals or other parts, a suitable pressure release shall be provided. Any inaccessible point that might need lubrication during operation shall be provided with a lubrication fitting extended to the side or other accessible points of the apparatus.

3.19 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates. Nomenclature shall be "TRUCK, FIREFIGHTING, AERIAL LADDER, 100 FOOT LENGTH, REAR MOUNTED."

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3.19.1 Transportation data plate. A transportation data plate shall be furnished by the contractor and securely affixed to the truck in a conspicuous location by the contractor with nonferrous screws or bolts of not less than 1/8 inch diameter. Nomenclature shall conform to the nomenclature on the unit identification plate. Transportation plate shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the truck showing the center of gravity shall be provided on the transportation plate.

3.19.2 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8 inch diameter.

3.20 Treatment and painting. Truck surfaces shall be cleaned and treated in accordance with MIL-T-704, type A. The exterior surfaces to be painted shall be primed with a zinc chromate epoxy applied to a dry film thickness of 1.5 - 2.5 mils. The primer used shall be compatible with the final paint system. The finish paint shall be polyurethane enamel or equal, applied to a dry film thickness of 1.8 - 2.2 mils. The finish color shall be lime yellow, gloss no. 13670 of FED-STD-595 and shall be applied to the exterior surface of the truck cab, chassis, and body. The roof and upper part of the cab to a line at the bottom of the windshield shall be painted white, color Chip No. 17886 of FED-STD-595. A white reflective glow type, horizontal "accent" band, 10 inches wide, shall be installed along the sides, the full length of the truck. The top of the band shall align approximately with the top of the wheel well opening. The interior surface of the equipment compartments shall be painted with spatter type paint to reduce problems in repainting due to chipping from equipment storage. Running boards, compartment tops, steps and turntable work platform shall be of aluminum color. The cab interior shall be painted two coats semigloss enamel conforming to TT-E-529 and Color Chip No. 24525 of Fed-STD-595.

3.20.1 Identification marking. Identification marking and data plates shall be in accordance with MIL-STD-1223 as specified for the Navy. All lettering and numerals shall be black, No. 17038, of FED-STD-595.

* 3.20.2 Rustproofing. The truck shall be rustproofed in accordance with FED-STD-297.

3.21 Servicing and adjusting. Prior to acceptance of the aerial ladder truck by the Government, the contractor shall service and adjust the aerial ladder truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.

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c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment.
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The aerial ladder truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.22 Workmanship.

3.22.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.22.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.22.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.22.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.22.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one truck when a first article is required (see 3.2 and 6.1). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. When first article is required, the first article shall be tested as specified in 4.4.1 through 4.4.6. The first article tests shall be performed by the contractor under the direction and in the presence of Government representatives. Failure to pass any phase of the required tests shall be cause for the Government to refuse acceptance of all trucks until corrective action has been taken.

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4.4.1 Operational test. The truck, completely equipped for firefighting and with a full crew, or equivalent deadweight, shall be tested as follows:

a. Ten miles over paved roads at maximum traffic speeds up to 55 mph.
b. Gradeability test to verify conformance to the requirements of 3.6a.
c. Acceleration test to verify conformance to the requirements of 3.6b and 3.6c.
d. Service brake test to verify conformance to the requirements of 3.6e.
e. Parking brake test to verify conformance to the requirements of 3.6f.

* 4.4.2 Aerial ladder tests. A structural strength test of the aerial ladder shall be conducted in accordance with NFPA 1904. In lieu of testing, an approved Third Party inspection certificate accompanied by the aerial ladder manufacturer's certification that the aerial ladder furnished has been previously tested to comply with NFPA 1904, is acceptable. In addition, an operational test of the aerial ladder shall be conducted. A complete cycle of aerial ladder operation shall be carried out after starting the engine, setting the outriggers, and transmitting power to the ladder. Elevating the ladder out of the bed, rotating ladder 90° and extending ladder to full specified height shall be completed smoothly and without undue vibration in not over 60 seconds. The ladder shall be retracted, the turntable rotation completed and the ladder lowered to its bed after which a thorough inspection shall be made of all moving parts with special attention being given to the security and adjustment of the ladder cables. The test shall include successful operation of all ladder controls, safety devices, auxiliary power system, and ground outrigger devices.

4.4.3 Ground ladder tests. Each ground ladder specified in 3.15 shall be tested in accordance with NFPA 1931, Article 130. In lieu of testing, inspection accompanied by ladder manufacturer's certification that the ladders have been factory tested and comply with NFPA 1931, is acceptable.

4.4.4 Engine and cooling system tests. A high and low temperature operating test shall be performed to verify engine and cooling system requirements of 3.5 and 3.8.6 respectively. Certification of high and low temperature operating requirements of engine and cooling system may be furnished in lieu of hot and cold room tests.

* 4.4.5 Ladder pipe tests. The aerial ladder and water system shall be tested for compliance to requirements specified in paragraphs 3.12 and 3.12.1.

4.4.6 Hydraulic system tests. The hydraulic system shall be checked for conformance to 3.14 through 3.14.5. With pump engaged and operating at recommended speed, check hydraulic pressure to see that it is within limits recommended by the manufacturer.

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4.4.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.5 Quality conformance tests. Each production truck shall be tested as specified in 4.4.2 through 4.4.5. Failure to conform satisfactorily to any phase of the tests shall be cause for rejection.

4.6 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packaged in accordance with the contractor's standard practice. When specified, equipment shall be preserved and packaged in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging as specified (see 6.1).

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
c. When vehicle shall be preserved and packaged in accordance with MIL-STD-281 with the level of preservation and packaging required (see 5.1).

6.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.3 First article. When a first article inspection is required (see 6.1), the truck will be tested and should be a first production item consisting of one complete truck or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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* 6.4 Subject term (key word) listing.

Diesel engine
Fire fighting vehicle
Fire protection
Fire pump

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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